

Research Article

Monitoring-Based Evaluation of Wind-Induced Vibration and Travel Comfort of Long-Span Suspension Bridge

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In this paper, evaluation of wind-induced vibration and travel comfort of the long-span suspension bridge were comprehensively conducted based on multisource monitoring data. The wind field distribution and turbulent characteristics during the normal and vortex-induced vibration (VIV) period were comparatively revealed. It reveals that the bridge experienced vertical VIV due to the long-duration wind with specific speed perpendicularly acting on the girder, which cannot be predicted by the turbulence intensity and gust factor. Meanwhile, dynamic response evolution, VIV lock-in effect, modal identification, and wavelet spectrum were further explored based on displacement and acceleration. The VIV frequency was consistent with a natural frequency of the bridge, whose mode can be determined by the deflection correlation heat map. The VIV was due to periodic vortex shedding generate aerodynamic forces, and the reaction of the structure vibration on vortex shedding can cause the vortex shedding frequency to be “locked” over a considerable range of wind speeds. According to driving visual safety and vibration tolerance, it is indicated that such VIV of the bridge may lead to the very discomfort for driving and pedestrian can tolerate short-term vibration in this period. Comfort evaluation for the bridge during the VIV should be further improved accuracy and reliability, which can contribute to emergency response to VIV situations. Note that a certain degree of discomfort may occur under normal vibration conditions, which raises doubts about the reasonableness of the limit value.

Keyword: comfort evaluation; lock-in effect; structural health monitoring; vortex-induced vibration; wavelet spectrum

1. Introduction

Modern bridges tend to complex-sharped and light-weight structure for long span and esthetic shape [1, 2]. The high flexibility of long-span suspension bridges greatly triggers a susceptibility to the excitations of vehicles and wind. Wind-induced vibration, such as vortex and flutter [3], gradually becomes prominent in the potential disasters of the long-span bridges with bluff shape. The wind effects related to long-span suspension bridges have been

extensively investigated through full aeroelastic tests and numerical simulations [4, 5]. Vortex-induced vibration (VIV) caused by periodically vortex shedding occurs at low wind speeds frequently, while its increased vibration amplitude affects fatigue performance of the bridge and comfort of the driving. For example, first-order vertical VIV with a maximum amplitude of 0.5 m occurred in Tokyo Bay Bridge under the wind with a speed of 17 m/s nearly perpendicular to the girder [6]. Significant VIV phenomenon is more prominent on a number of long-span bridges,

including Storebælt Suspension Bridge in Denmark [7], Humen Bridge in China [8], Volgograd Bridge in Russia [9], and Yi Sun-sin Bridge in Korea [10]. Bridge collapse accidents such as Tacoma Bridge may not happen again [11], but research on the mechanism of VIV and vortex shedding characteristics of the bridge girder is still relatively defective.

As VIV occurs frequently in many long-span bridges, wind tunnel tests and numerical model simulation are extensively conducted to explore the VIV effect and mechanism. The VIV phenomena of suspension bridges have been experimentally studied based on advanced wind tunnel testing technique [12, 13], including accurate wind field simulation, model design method, and particle image velocimetry. The inflow velocity field, aerodynamic force, time-frequency evolutionary characteristics of different types of girders, and bridge components under VIV have been tested comprehensively [14, 15]. Note that the experimental results are limited by the size of scale model and wind tunnel, as well as high cost. Numerical model simulation, such as computational fluid dynamics (CFD), is potential in the mechanism research of wind-induced vibration. According to different simulation methods of fluid and solid domain, local flow field, vortex shedding, and fluid-structure interaction of bridge decks, cables, and towers during VIV have been simulated [16, 17]. Nevertheless, the accuracy and reliability of the simulation results are dependent on turbulence model, computational and calculation error with time-consuming.

As obtaining actual wind field characteristics and wind-induced vibration response directly, field measurement and health monitoring system can achieve prototype tests of VIV on a bridge [18]. Wind-induced behavior and performance of bridges are continuously explored in conjunction with measured dynamic responses from various aspects [19, 20], including strain, displacement, and acceleration. The continuous emergence of advanced methods based on modal variability, local correlation model, frequency cointegration, time-frequency local transformation, and deep learning has promoted a more accurate evaluation of wind-related behavior in bridges [21, 22]. Based on monitoring data, studies on the relationship between the wind field characteristics and structural modal parameters have been carried out [23, 24]. Li et al. [25] measured the vortex-induced oscillation events based on diverse sensors and investigated the evolution of flow around the bridge deck, lock-in state, and vortex shedding. Yang et al. [26] conducted wind tunnel tests on a single box girder at different angles of attack. The results show that the VIV of the single box girder is completely suppressed at positive angles of attack, and the self-emission flow control can significantly reduce the locking region at negative angles of attack. Some measured results show that the vertical VIV mostly occurs in the main beam when the wind speed is small, the turbulence is low, and the wind direction is basically perpendicular to the axis of the bridge [27]. Based on the monitoring data, many control methods for VIV have been truly validated and practically applied [28]. Besides, a series of investigations about the prediction and prevention of VIV have been conducted, where the human comfort of vehicle riding in the VIV

phenomenon can also be evaluated [29–31]. Hence, the prototype monitoring data make it feasible to rapidly capture and evaluate the VIV behavior during bridge operation. Nowadays, the design is believed to have the ability to avoid VIV, and a number of bridges, such as Yingwuzhou Bridge, still suffer from severe vibration under traffic volume. The existing SHM provides a good basis for revealing the mechanism and characteristics of such vibration response.

In this paper, evaluation of wind-induced vibration, especially VIV, and travel comfort of the long-span suspension bridge are comprehensively conducted based on multisource monitoring data. First, the prototype bridge, monitoring system, and structural dynamic characteristic are described in detail. Second, the distribution of wind speed and direction and turbulent characteristics during the normal and VIV period are comparatively analyzed. Next, dynamic response evolution, VIV lock-in effect, modal identification, and wavelet spectrum are further explored based on displacement and acceleration. Finally, the comfort for driving and pedestrian has been evaluated according to the driving visual safety and vibration tolerance.

2. Descriptions of Prototype Bridge and Monitoring System

2.1. Bridge Configuration. The Yingwuzhou Yangtze River Bridge (YYRB), located at Wuhan, China, is the world's first triple-tower four-span suspension bridges, as shown in Figure 1. The main bridge is 2150 m with a span arrangement of $(225 + 2 \times 850 + 225)$ m, as shown in Figure 2(a). The opening box girder in Figure 2(b) is a steel-concrete composite structure with a maximum height of 3.0 m, consisting of two steel main beams, steel cross beams, a concrete bridge deck, and two wind fairings. To improve the wind resistance performance [32], wind fairings with an angle of 95° are designed and welded on the two outer sides of the girder, making its width of up to 38 m. The standard segment of the girder is 15 m, and the center distance of those two main beams is 31.2 m. The main beams have an I-shaped cross-section with a total height of 2.423 m, where the top plates have a fixed specification and other components vary at different positions. Cross beams with an I-shaped cross-section and a height of 2.735 m at the mid span are installed between the two main beams with a longitudinal spacing of 3.0 m in standard segment. Besides, I-shaped longitudinal beams are bolted to the webs of the transverse beams are for strengthening. The bridge deck adopts prefabricated panels with cast-in-place joints and is connected to the girder through shear studs. The thickness of bridge deck and pavement is 0.2 and 0.065 m, respectively.

The middle tower is a steel-concrete hybrid structure, while the side towers are concrete structures. The connection between the tower and the girder is a semifloating structure. Movable ends are set at the two side towers, and fixed supports are set at the middle tower and two side piers. Considering that the two main spans have a large turning angle at the middle tower, a 5.5 m long transition beam is installed between the two main spans at the top of the crossbeam of the middle tower to improve the driving



FIGURE 1: Yingwuzhou Yangtze River bridge.

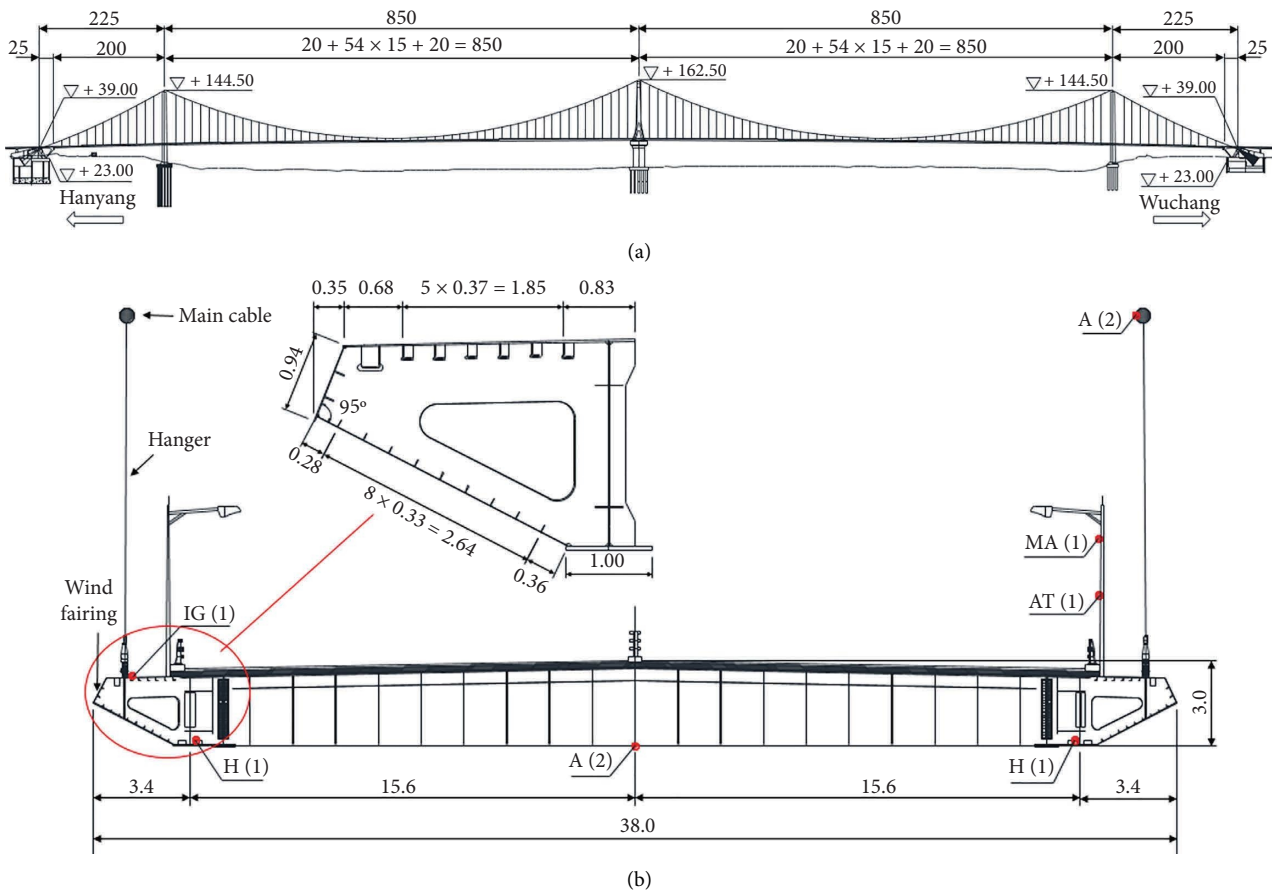
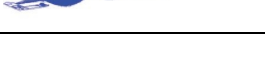


FIGURE 2: Bridge configuration and cross section of opening box girder. (a) Configuration of the YYRB. (b) Cross section of opening box girder.

conditions. To resist lateral wind and longitudinal/transverse seismic forces, the girder is equipped with longitudinal and transverse seismic blocks at the middle tower and side towers, while the two side piers are equipped with longitudinal seismic supports and transverse seismic blocks. Besides, two consecutive main cables are used to support the girder, whose diameter is about 700 mm and horizontal spacing is 36 m. The aspect ratio of the main cables is 1/9. Both main cables consist of 114 high-strength steel wire strands. High-strength steel wire suspenders with a longitudinal spacing of 15 m are adopted to connect the girder and main cables.

2.2. Dynamic Characteristic of Bridge. The dynamic characteristics of YYRB have been tested via the completion test [33], where the consistency of test and design confirms the compliance with design and safety operation requirements. Table 1 lists some dynamic characteristic information, including frequency, damper ration, and mode shape [34]. The measured frequency of first-order vertical symmetrical bending vibration and symmetrical torsional vibration is 0.103 and 0.215 Hz. Hence, the first-order torsion-bending frequency ratio is 2.09. Besides, the damper ration of the bridge is within the range of 0.19% to 0.70%.

TABLE 1: Dynamic characteristic of the YYRB.

Model number	Natural frequency		Damper ration (%)	Mode shape	Vibrational pattern
	Design value	Test value			
1	0.097	0.103	0.19	1 st order vertical antisymmetric bending	
2	0.128	0.135	0.41	2 nd order vertical antisymmetric bending	
3	0.159	0.165	0.70	3 rd order vertical symmetrical bending	
4	0.181	0.188	0.61	3 rd order vertical antisymmetric bending	
5	0.200	0.208	0.34	1 st order asymmetric torsional bending	
6	0.205	0.215	0.39	2 nd order symmetric torsional bending	
7	0.208	0.228	—	3 rd order asymmetric torsional bending	
8	0.214	0.248	—	4 th order vertical antisymmetric bending	

2.3. Monitoring System. A monitoring system consisting of 223 sensors has been implemented to measure the structural responses and changes since the completion of the YYRB [35]. Those sensors were installed in a total of 25 monitoring sections (labeled as S1 to S25) along the bridge, including the two ends of the girder. Figure 3 illustrates the arrangement of the sensors. To measure the wind and temperature, two mechanical anemometers and 83 atmospheric thermometers were applied. The two mechanical anemometers were installed at the mid-span of the left main span (i.e., in S9) and at the top of the middle tower (i.e., in S13), respectively. Lateral and longitudinal acceleration sensors were installed in the section of S9, S11, S12, S14, and S17. Besides, a longitudinal acceleration and a three-axis acceleration sensor were, respectively, installed on the tower top and cushion cap. Inclinerometers were used to capture the incline of the three towers and the rotation of the four mid-spans. GPS receivers were used to monitor the displacement at the mid-span of two main spans. Meanwhile, magnetostrictive displacement sensors were equipped to adopt the displacement of each support. Each monitoring section was equipped with a static pressure leveling instrument to measure the deflection of the girder. The modal and performance of the sensors are listed in Table 2. The sampling frequency of accelerometers is set to 20 Hz, and the sampling frequency of other sensors is set to 0.25 Hz.

The video monitoring system captures a sequence of images of the beam in real time using cameras and analyzes these images using video displacement tracking methods. These methods can accurately track the displacement of

surface markers or feature points on the structure, allowing the calculation of the deformation and displacement of the structure under applied loads [36]. The steps for using the video monitoring system are as follows:

1. Video monitoring was used to capture the girder movement, which was installed on the cushion cap of the left side tower. An air vent on the wind fairing was set as the feature point of the girder for video tracking. The video was converted into image every 25 frames.
2. The template matching technique is applied to the obtained pictures. The initial state of the girder is selected as the template, and the displacement of the girder can be obtained after coordinate conversion by comparing the relative displacement of the subsequent pictures and the template.
3. The camera system starts capturing continuous video images to record the deformation process of the girder under load. These images will be used as the basis for subsequent analysis.
4. Image processing software is utilized for image alignment, feature point extraction, and tracking to calculate the displacement of each marker point at different time points. The deformation and displacement data of the girder are obtained by analyzing the changes between different image frames. Sharpening processing and edge recognition algorithms were used for target position. Hence, the dynamic information can be achieved via the time history of position.

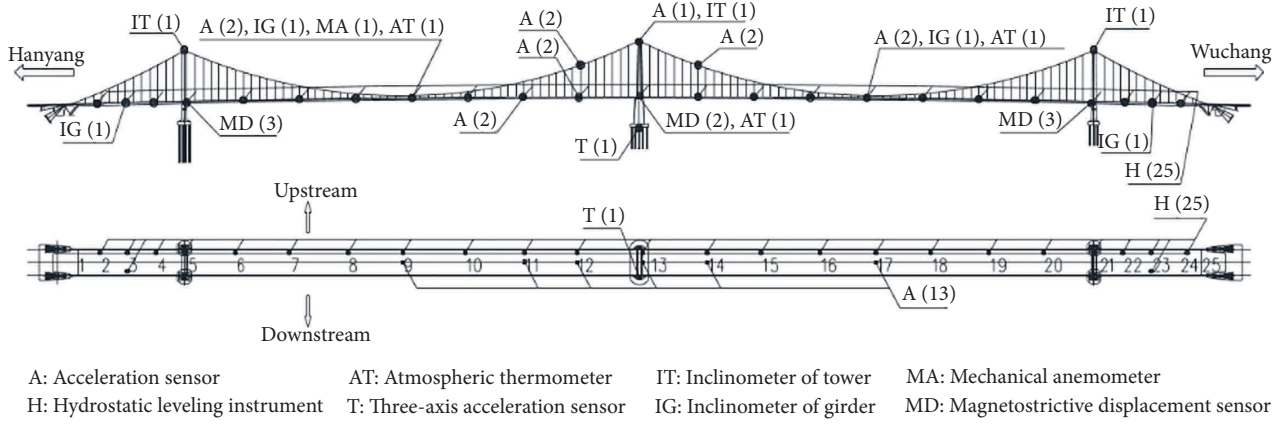


FIGURE 3: Arrangement of monitoring system (unit: m).

TABLE 2: Sensors in the monitoring system.

Category	Model	Precision	Number
Mechanical anemometer	DEL 5011	$\pm 0.4 \text{ m/s}$ and $\pm 2^\circ$	2
Ambient temperature sensor	LTM9901M	$\pm 0.5^\circ\text{C}$	3
Structural temperature sensor	LTM8877	$\pm 0.5^\circ\text{C}$	80
Pressure leveling instrument	EJA110A	0.075%	25
Inclinometer	LCF100	$< 0.02^\circ$	7
Displacement sensor/LVDT	KYDM	$\pm 0.3 \text{ mm}$	8
GPS receiver	N71M	$2.5 + 0.5$ and $5 + 0.5 \text{ ppm}$	4
Vibration wire strain sensor	BGK-4200/4000	$3.75 \mu\epsilon$	80
Single-axis acceleration sensor	GT-02	$\pm 2.5 \text{ V/g}$	13
Three-axis acceleration sensor	GT41-3	$\pm 2.5 \text{ V/g}$	1

3. Wind Field Characteristic During Special Vibration

Obvious vibration of the girder of the YYRB was experienced during the period of 12:00 to 13:00 and 16:10 to 17:10 on April 16, 2020. The monitoring system has fully recorded the environmental information and structural responses. Figure 4 illustrates the wind speed and direction for that day to investigate the wind field characteristic. The period from 7:00 to 19:00 was identified for concern as the wind direction basically unchanged with considerable speed. It is observed that the wind speed increased rapidly from around 7:00 and did not drop below the speed at 7:00 for the first time. The average wind speed and maximum wind speed during the period were 7.45 and 9.33 m/s, respectively, which was far below than the design wind speed and minimum flutter wind speed of the girder (i.e., 30.5 and 46.1 m/s). According to Figure 4, the wind direction kept within 186° – 216° during this special period. Figure 5 further illustrates the distribution of wind during the normal vibration and special vibration of the bridge. It is found that the wind direction was usually at a large angle to the girder, which almost perpendicularly acted on the girder during the special vibration. According to the girder characteristic, the critical wind speed of VIV was calculated according to the standard, which was 7.83 m/s for vertical VIV and 10.5 m/s for torsional VIV. Hence, it is indicated that this special vibration with limited amplitude is attributed to vertical VIV. Note

that no significant vibration of the bridge has been observed, although wind in this magnitude often occurs at the bridge site. Therefore, the bridge experienced VIV due to the long-duration wind with specific speed perpendicularly acting on the girder.

To explore the turbulent characteristics of the wind field, turbulence parameters, i.e., turbulence intensity and gust factor [37], are further calculated according to equations (1) and (2).

$$I = \frac{\sigma_U}{U}, \quad (1)$$

$$G = \frac{\max(u(t_g))}{U}, \quad (2)$$

where I is the turbulence intensity; σ_U and U are the standard deviation and mean value of the fluctuating wind speed, respectively; G is the gust factor; t_g is gusty duration, which is set as 4 s here. From equation (1), it can be seen that the standard deviation σ_U of the wind speed tends to be large when the wind speed changes considerably in a short period of time. At this time, the turbulence intensity I will also exhibit a higher level. Similarly, the maximum value of the mean wind speed at time t_g during the gust is much larger than the mean wind speed during the gust. The gust factor G and the turbulence intensity I reflect a similar pattern; i.e., they exhibit higher values when the wind speed changes considerably over a short period of time.

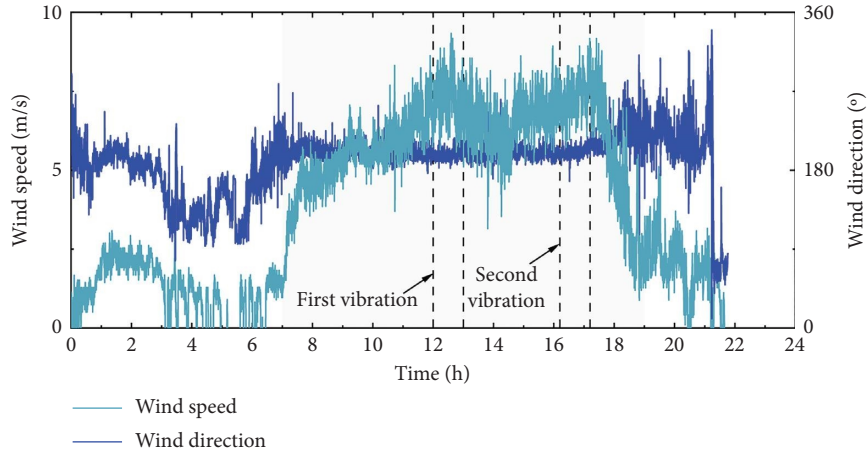


FIGURE 4: Time history of wind speed and direction.

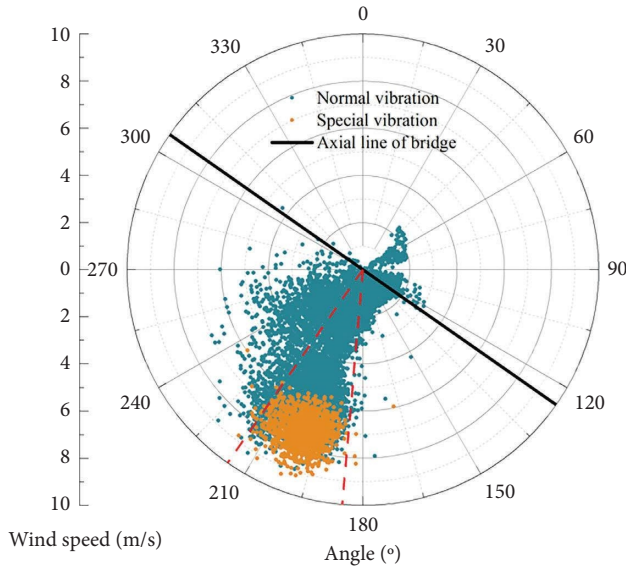


FIGURE 5: Distribution of wind speed and direction.

Figure 6(a) shows the time history of turbulence intensity during the period of 7:00 to 19:00 on April 16, 2020, where turbulence intensity during the VIV and normal vibration of bridge are most between 7.0% and 8.0%. Further, the turbulence intensity ranged from 7.0% to 9.0% during the first VIV. Besides, the turbulence intensity was distributed in the range of 6.0% to 8.0% during the second VIV. Apparently, the VIV cannot be predicted by turbulence intensity. Figure 6(b) further statistically analyzes the gust factor and wind speed. The gust factor parameters were distributed in the range from 1.18 to 1.62, which basically overlapped with some during normal vibration. No obvious characteristic was found from the distribution of gust factors under normal and special vibration directly. Hence, the turbulence intensity and gust factor cannot be utilized as indexes to predict the occurrence of the VIV.

4. Structural Dynamic Response Under VIV

ANSYS Fluent is employed to model the two-dimensional wind-girder interaction. The turbulence is characterized by the RNG $k - \epsilon$ model. The angle of attack of the wind is 0° in the flow field, i.e., horizontal wind. This setup simplifies the calculation and effectively analyzes the dynamic response of the girder under the horizontal wind field. VIV is a limited-amplitude forced vibration with self-excited properties, which commonly occurs in girders, towers, and cables. Figure 7 shows the local wind field when the wind passes through the girder of the YYRB. The inflow is separated by the wind fairing and formed vortices on the surface of the girder. Periodic shedding of the vortices can generate aerodynamic forces and moment on the girder and induce VIV when the shedding frequency approaches a certain natural frequency of the structure [38, 39]. In the same wind environment, the shape of the girder is significant to the occurrence of the VIV. Although smaller angle improves the VIV performance [40], obtuse wind fairings have adopted by the YYRB for opening shape with small-angle end sensitive to torsional VIV. The vortex shedding frequency can be calculated by equation (3):

$$f_s = \frac{S_t V}{H}, \quad (3)$$

where S_t is the Strouhal number, which is 0.094 according to the CFD simulation; H is the feature size of the windward surface and equal to 3.0 m; and V is the inflow speed, which can be set to 7.2 m/s.

According to equation (3), the vortex shedding frequency of the YYRB is 0.23 Hz, which consists of the frequency of the fourth-order vertical antisymmetric bending (0.248 Hz in Table 2). Hence, this further confirms the occurrence of significant vertical VIV in the girder of the YYRB.

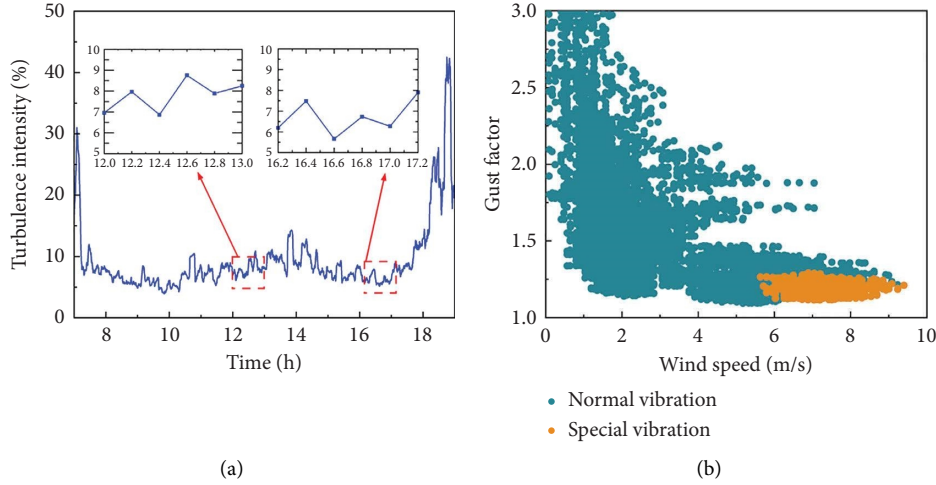


FIGURE 6: Comparison of turbulence parameters for normal and special vibration. (a) Time history of turbulence intensity. (b) Gust factor.

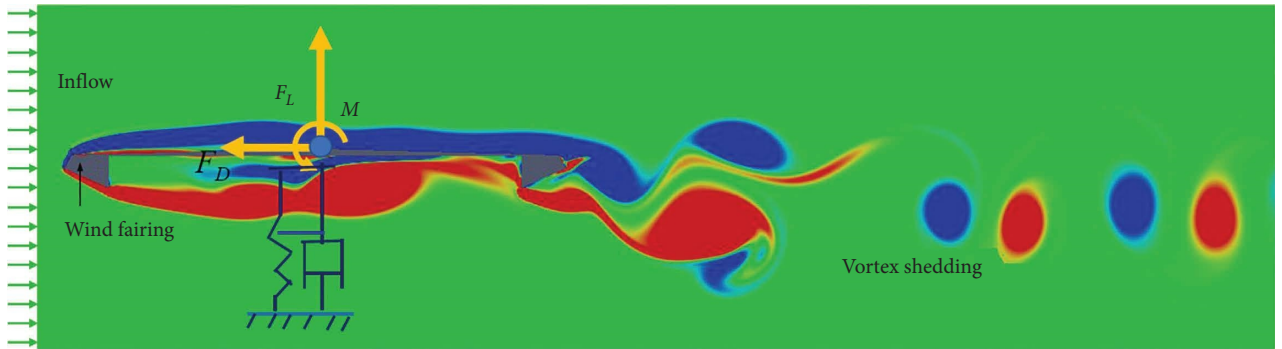


FIGURE 7: Schematic of local flow field.

4.1. Displacement Response and Lock-In Effect. Figure 8(a) presents the displacement time histories of all the 25 monitoring sections of the girder, which were obtained by the pressure leveling instruments. Note that all measured data used here were filtered to exclude the low-frequency part caused by temperature [41]. All the displacements significantly increased in the period of 12:00 to 13:00 and 16:10 to 17:10, confirming VIV state of the girder during these two periods. It is found that the S8 (at the 3/8 span of the left main girder) had the maximum deflection, although S9 also had significant deflection. The maximum deflection amplitude was 0.55 m under the VIV. Figure 8(b) presents the motion displacement of the girder feature point by video tracking. It is found that the vertical vibration of the feature point was approximately sinusoidal motion and whose period was about 4.1 s. Hence, the vibration frequency was approximately 0.244 Hz.

Figure 9 further statistically analyzes the deflection and wind speed, where significant vibration amplitude occurred in a region. This is attributed to the lock-in effect [42]. Vortex shedding frequency commonly increases with wind speed. The VIV occurred when the vortex shedding frequency reached a natural frequency of the bridge, 0.244 Hz here. However, the vibration of the bridge reacted with vortex shedding, causing the vortex shedding frequency to

be “locked” by the natural frequency of the structure over a considerable range of wind speeds. Although the wind speed increased, the vortex vibration frequency remained unchanged in the lock-in range. The wind speed interval in which the vortex shedding frequency remains constant is called the locked wind speed interval. As the continuous increase in wind led to an increase in air damping, the vortex shedding frequency was unlocked and the VIV stopped [43]. Hence, the continuous significant vibration would decrease.

To determine the mode shape of the girder under the VIV, the deflection correlation of monitoring sections of the girder is calculated, as shown in Figure 10. The deflection correlation of the two mid-span sections (i.e., S9 and S17) was negative according to the scatter plot in Figure 10(a), indicating that the mode shape of the entire girder should be in antisymmetric during the VIV. Figure 10(b) further illustrates the correlation heat-map of all monitoring sections of the left main span. The S8 was set as reference for having the maximum deflection and small possibility of antisymmetric point. The deflection of S9 (at 4/8 span) was negatively correlated with that of S8 (at 3/8 span), indicating that the mid-span had an antisymmetric point. Hence, the mode shape of the single main span should be antisymmetric during the VIV. Furthermore, S7 was positively correlated with the S8 and negatively correlated with S6, indicating that

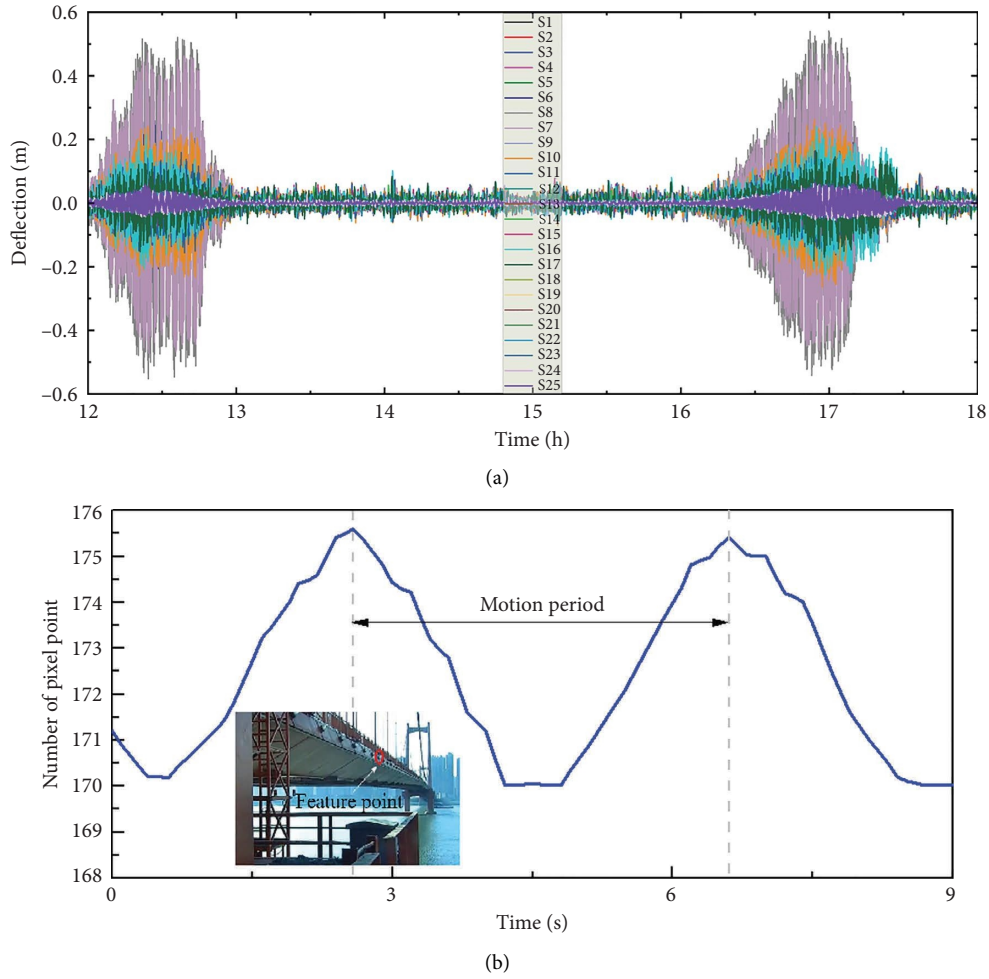


FIGURE 8: Displacement time histories of the girder. (a) Deflection time histories of the girder. (b) Motion displacement of girder feature point by video tracking.

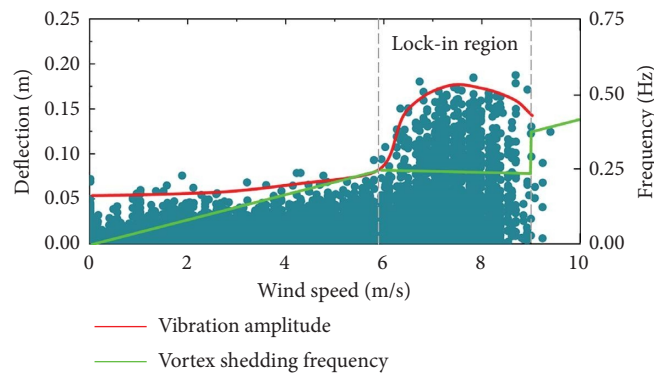


FIGURE 9: Displacement time histories of the girder.

the half span can generate a waveform. In addition, the actual natural frequency calculated by video inspection and demonstrated by the measured vibration data is 0.244 Hz, which is closer to the frequency of the fourth-order vertical antisymmetric bending. Therefore, the mode shape of the entire girder should be fourth-order vertical symmetrical bending.

Figure 11 illustrates the displacement response of the towers, which was monitored by using GPS. The spatial position changes of the main tower were relatively concentrated, and the displacement distribution in the longitudinal and transversal directions of the bridge was relatively uniform. No significant displacement concentration effect was observed in that day, as shown in Figure 11(a). Besides,

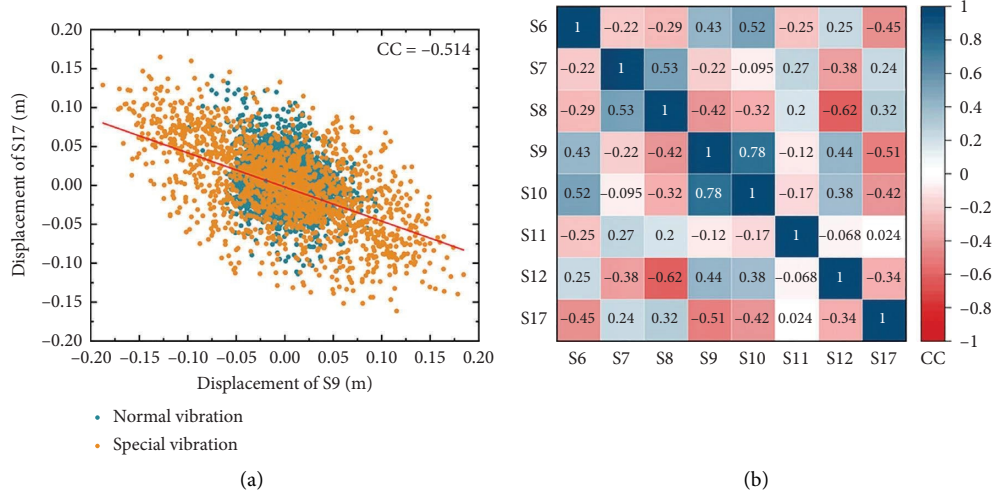


FIGURE 10: Deflection correlation of monitoring sections of the girder. (a) Deflection scatter plot. (b) Deflection correlation heat-map.

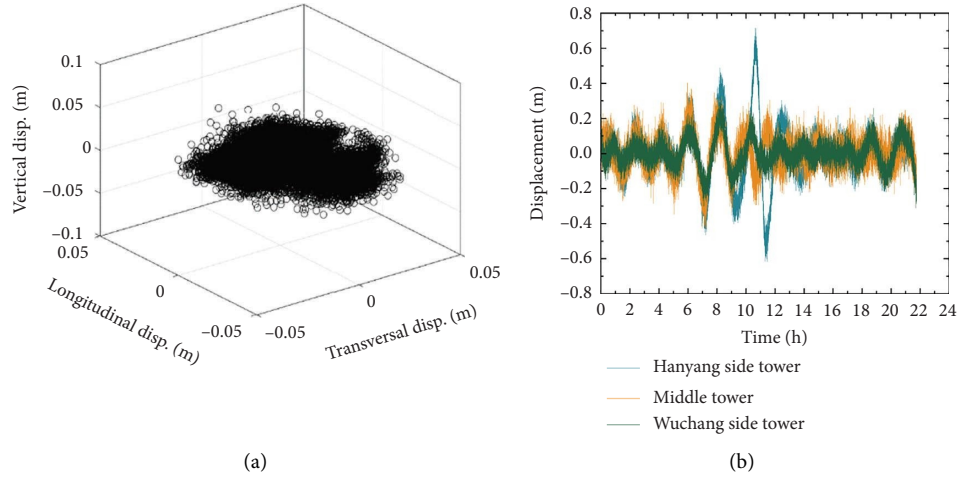


FIGURE 11: Displacement response of the towers during the VIV. (a) Spatial position of main tower. (b) Transversal displacement of three towers.

the displacement had slight response during the VIV of the girder. Hence, the towers did not experience VIV, and the VIV of the girder had little impact on the towers. Figure 11(b) presents the transversal displacement of three towers. It can be seen that there is no significant increase in displacement of the bridge tower during VIV. Portal bridge towers usually have strong transverse stiffness. The vibration response in the transverse direction can be effectively suppressed due to the stronger girders and tower columns. If the tower in the transverse direction has sufficient rigidity, the amplitude of this vibration will be limited, which will be manifested as a small transverse displacement.

Figure 12 illustrates the displacement response of the side support, which was monitored by using magnetostrictive displacement sensors. The trend of longitudinal displacement was consistent with the variation of temperature, and the numerous small fluctuations were caused by excitations of vehicle and wind [44], as shown in

Figure 12(a). Figure 12(b) shows the longitudinal displacement time history of the side support after filtering temperature influence by wavelet decomposition. Longitudinal displacement exhibits minor perturbations but is not significant under the influence of VIV. The maximum value of longitudinal displacement is 2.11 mm, which can be considered as the longitudinal displacement of the bridge is almost unaffected by VIV compared to the displacement of 15 mm in a day caused by temperature changes. However, two obvious transversal slides of the support occurred under these two VIVs, as shown in Figure 12(c). The support rapidly slid a large displacement at the beginning of the first VIV and did not reset automatically at the end, while slowly returned to its original position after the second rapid slide at the end of the second VIV. This phenomenon is probably the VIV promotes the start of two slides and the wind perpendicularly acting on the bridge provides sufficient lateral force.

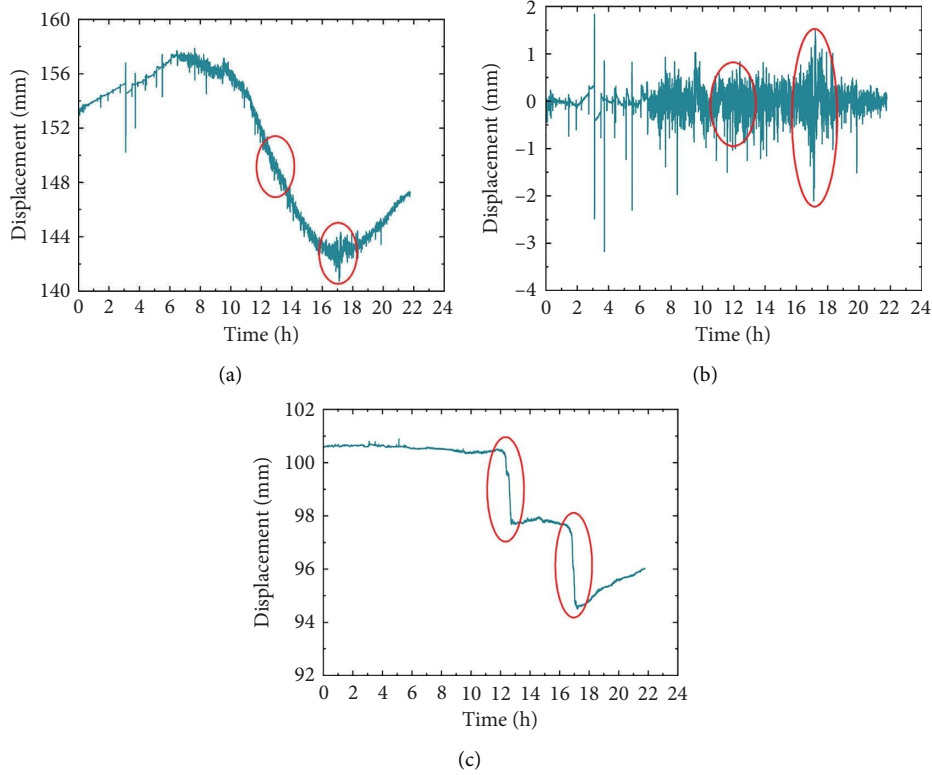


FIGURE 12: Displacement of side support. (a) Longitudinal displacement. (b) Longitudinal displacement without temperature effect. (c) Transversal displacement.

4.2. Acceleration Response and Frequency Spectrum. It can be seen from the analysis of the 25 deflection monitoring data as shown in Section 4.1 that the deflection responses at S9 and S17 are the most pronounced. Therefore, the acceleration responses at S9 and S17 are focused to be analyzed in this section. Figure 13 depicts the acceleration at the two mid-span sections of the girder, where some significant amplitudes indicate the existence of VIV. It is worth noting that the transversal vibration acceleration at the same monitoring point was significantly greater than the vertical vibration acceleration. Larger transversal stiffness may contribute mainly for this. Compared with S9 with larger deflection as shown in Figure 13(a), the acceleration at the S17 had a more prominent response to the VIV. It can be inferred that the structure with higher stiffness is more sensitive to external excitation and has larger acceleration response with small deflection. According to Figure 13(b), the transversal acceleration of the main cable was also affected. Considering the lack of VIV for the main cable under such wind excitation, it is due to the influence of the main cable. Frequency spectrum analysis [45] was conducted on the vertical acceleration of the mid-span under normal and special vibration, as shown in Figure 14. Energy was concentrated around the frequency of 0.2436 Hz during the VIV as shown in Figure 14(a), which was close to the frequency (i.e., 0.248 Hz) corresponding to the fourth-order vertical antisymmetric bending. For comparison, the power spectral density under normal vibration was also calculated, as shown in Figure 14(b). No single energy concentration was

found, and the typical modes can be well identified for clear frequency spectra.

To further investigate transients in nonstationary process, wavelet analysis that is a method of transforming time-sequential data-to-data on a time-frequency plane was conducted on the acceleration time history of the girder. The calculation formula of continuous wavelet transform (CWT) that is defined as the convolution of data with a scaled and translated version is as follows [46]:

$$W(a, \tau) = \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t - \tau}{a} \right) dt, \quad (4)$$

where W is the basis function; a and τ represent the scale parameter and shift parameter; and ψ^* is the complex conjugate of wavelet function.

Figure 15 presents the magnitude scalogram and corresponding scale-averaged wavelet spectrum. The cone of influence was considered as shown in Figure 15(a) and included the dashed while and the shaded region from the edge of the line to the frequency and time axis, in which information should be treated as suspect due to the potential for edge effects. As shown in Figure 15(a), the VIV can be easily located according to the magnitude, which can be used for monitoring and early warning by integrating visual recognition. The scale-averaged wavelet spectrum as shown in Figure 15(b) reflects the distribution of wavelet power spectrum along the time direction. However, no significant energy concentration occurred during the VIV, which cannot be used to characterize the VIV phenomenon.

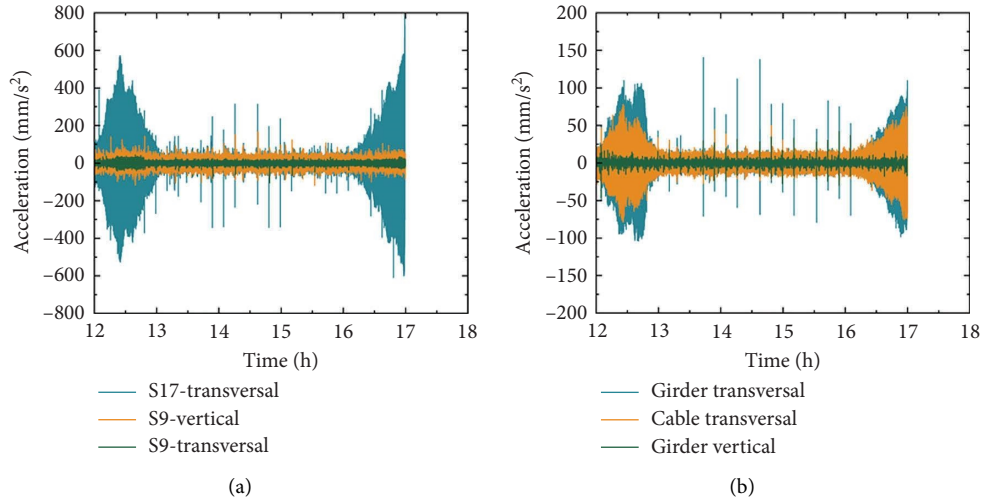


FIGURE 13: Acceleration response of the bridge for VIV. (a) Acceleration of the two mid-span. (b) Acceleration of the girder and cable.

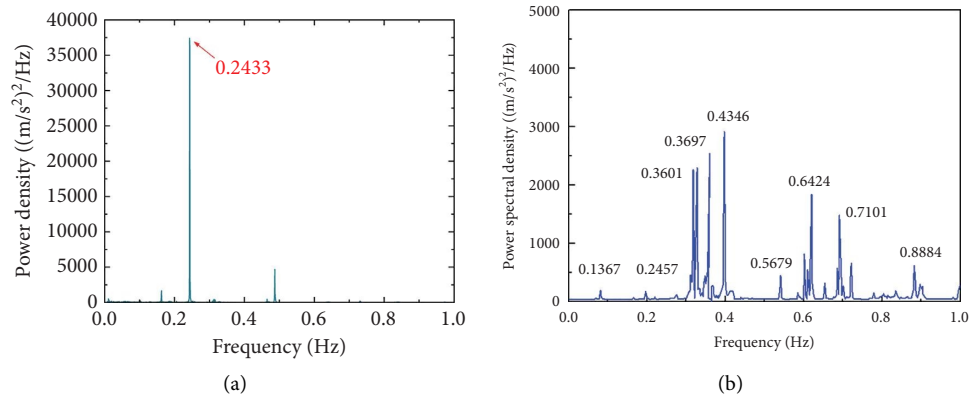


FIGURE 14: Frequency spectrum analysis of the girder during normal and special vibration. (a) Frequency spectrum during VIV. (b) Frequency spectrum during normal vibration.

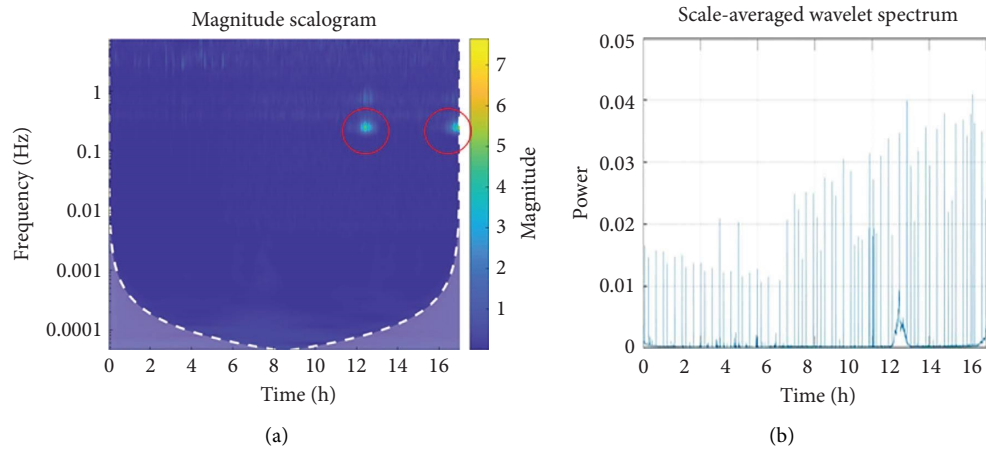


FIGURE 15: Acceleration time-frequency spectrum of continuous wavelet transform. (a) Magnitude scalogram analysis. (b) Scale-averaged wavelet spectrum.

5. Evaluation of Travel Comfort for VIV

A bridge that sways like waves under the VIV can directly lead to dizziness and other physical discomfort for travelers and even cause great concern for structure safety and social panic. Hence, the evaluation of travel comfort should be conducted to provide support and emergency plans for handling VIV situations.

5.1. Comfort Evaluation for Driving. Driving will be exacerbated by the VIV of the bridge, whose comfort level is mainly affected by the amplitude, frequency, velocity, and acceleration of bridge vibrations. According to displacement monitoring, the bridge performs simple harmonic vibration of a certain mode for during the VIV and each point of the bridge moves approximately in sine or cosine motion. The VIV amplitude (A_v) should not exceed 1 m/s^2 according to the general experience. Thus, the vibration amplitude limit (A_1) can be computed according to the VIV frequency (f_v), as shown in equation (5) and Figure 16.

$$A_1 = \frac{A_v}{4\pi^2 f_v^2}. \quad (5)$$

Since the VIV frequency of the YYRB was measured to be 0.244 Hz , the vibration amplitude limit should be 0.425 m according to equation (5). As the maximum deflection of the mid-span was 0.19 m , no vibration discomfort should exist. However, the $3/8$ span with a maximum deflection of 0.55 m had exceed the limit. Hence, it may lead to the discomfort for driving according to this preliminary evaluation.

Another vibration amplitude limit (A_2) has been recommended by some standards, as shown in equation (6) and Figure 16. Thus, the vibration amplitude limit should be 0.164 m . According to this strict limit, it may lead to the very discomfort for driving under the VIV according to this preliminary evaluation. Note that a certain degree of discomfort may occur under normal vibration conditions, which raises doubts about the reasonableness of the limit value.

$$A_2 = \frac{0.04}{f_v}. \quad (6)$$

Besides, from this perspective, another vibration amplitude limit (A_3) has been proposed. It requires that the amplitude should be limited to 0.35 m when the VIV mode is no less than third order, as depicted in Figure 16. Compared the deflection with the limit, the girder satisfied the comfort requirement under normal vibration and was on the discomfort for driving under the VIV according to this preliminary evaluation.

5.2. Comfort Evaluation for Pedestrian. Currently, the main evaluation indexes for comfort are Spelling index and Dyckman index, in which Spelling index is mainly used to evaluate the comfort of drivers and passengers while Dyckman index (i.e., vibration sensitivity K) is used to evaluate the comfort of pedestrians [47]. Here, the main

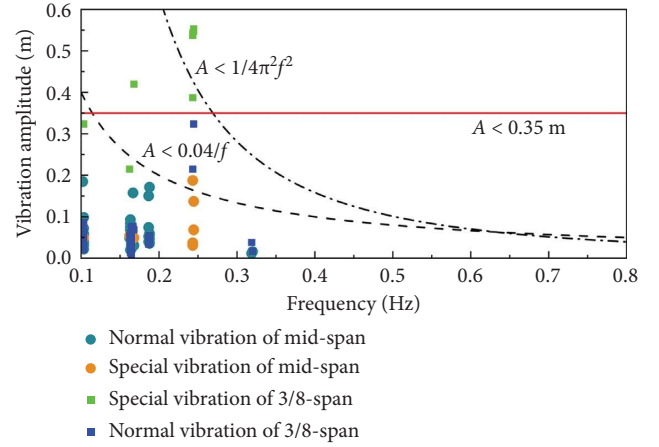


FIGURE 16: Vibration comfort limit with frequency variation.

focus was on the comfort situation for pedestrians, so the Dyckman metric was chosen to evaluate it. The calculation formulas of K for pedestrian passing by the girder are as follows:

For vertical vibration,

$$K = \begin{cases} Af^2, & f < 5 \text{ Hz}, \\ Af, & 5 \text{ Hz} \leq f \leq 40 \text{ Hz}, \\ 200A, & f > 40 \text{ Hz}. \end{cases} \quad (7)$$

For transversal vibration,

$$K = \begin{cases} 2Af^2, & f < 2 \text{ Hz}, \\ 4Af, & 2 \text{ Hz} \leq f \leq 25 \text{ Hz}, \\ 100A, & f > 25 \text{ Hz}, \end{cases} \quad (8)$$

where A is the vibration amplitude in mm and f is the vibration frequency in Hz . The vibration sensitivity range based on Dyckman index is as follows [48, 49]: $K=0.1$ is lower limit at which vibration can be felt; $K=1$ is the limit for tolerating vibrations for any length of time; $K=10$ is the limit for tolerating short-term vibration; and $K=100$ is the upper limit of excessive fatigue caused by vibration for ordinary people.

During the VIV, the measured vibration frequency was equal to 0.244 Hz , and the maximum vertical vibration amplitude was 550 mm . According to equation (7), the calculated K of vertical vibration is 32.74 , meaning humans can tolerate short-term vibration in this period. Meanwhile, the maximum transversal vibration amplitude was 32.1 mm during the VIV. According to equation (8), the calculated K of transversal vibration is 3.82 , meaning humans can tolerate vibrations for any length of time. Thus, the vibration comfort evaluation was poor due to the vertical vibration under the VIV, and pedestrians should be restricted from crossing the bridge during this period.

6. Conclusions

This paper presents a comprehensive investigation evaluation of wind-induced vibration and travel comfort of a long-

span suspension bridge based on multisource monitoring data. The wind field distribution and turbulent characteristics during the normal and VIV period were comparatively revealed. Meanwhile, dynamic response evolution, VIV lock-in effect, modal identification, and wavelet spectrum were further explored. According to the driving visual safety and vibration tolerance, the comfort for driving and pedestrian had been further evaluated. Based on the above research results, the following conclusions can be drawn:

1. Comparative analysis of wind field and turbulent characteristics during the normal and special vibration reveals that the YYRB experienced two vertical VIVs under the average wind speed of 7.45 m/s. The mechanism of VIV is the long-duration wind with specific speed perpendicularly acting on the girder and making the vortex shedding equal to a natural frequency of the bridge (about 0.244 Hz). Besides, turbulence intensity ranged from 6.0% to 9.0% and gust factor was distributed within 1.18 to 1.62 during the VIV, which cannot be VIV predictor for basically overlapping with normal vibration.
2. Dynamic response evolution of the bridge under normal vibration and VIV had been explored. The vertical displacement and transversal acceleration of the girder were more sensitive to the vertical VIV for larger lateral girder stiffness, and the maximum deflection amplitude was 0.55 m near the mid-span of the left main span. Although no VIV occurred in the cable and tower, the cable was also affected by the vertical VIV of the girder and the tower had no influence. Mode shape was successively identified based on a method of the deflection correlation heat-map.
3. The VIV was due to periodic vortex shedding generate aerodynamic forces, and the reaction of the structure vibration on vortex shedding can cause the vortex shedding frequency to be “locked” over a considerable range of wind speeds. The vortex shedding frequency was unlocked, and the VIV stopped as the continuous increase in wind led to an increase in air damping. The VIV can be easily located according to the magnitude scalogram from frequency spectrum analysis, which can be a potential method for monitoring and early warning by integrating visual recognition.
4. According to driving visual safety and vibration tolerance, the VIV of the YYRB may lead to the very discomfort for driving, and the pedestrian comfort was poor in this period. The vibration amplitude limit $A_2 = 0.04/f_v$ is unreasonable as a certain degree of discomfort may occur under normal vibration conditions. Comfort evaluation for the bridge during the VIV should be further improved accuracy and reliability, which can contribute to emergency response to VIV situations.

This study explores the VIV characteristics of YYRB through the monitoring data. The main innovations are as follows: (1) Amplitude variations of girders during the lock-in intervals are explored. (2) Deflection correlation

heat-map of different monitored cross-sections is used to reflect the vibration modes of the bridge. (3) CWTs of girder acceleration data are performed to obtain the magnitude scalogram figure to represent the vortex vibration frequency, suitable for monitoring and early warning by integrating visual recognition. (4) Driving comfort index is proposed by comparing the girder vibration amplitude with empirical indexes, where the rationality of different vibration amplitude limits is first evaluated. However, there are still some limitations in this study. Due to the limited number of wind speed and direction sensors, the consideration of vortex vibration angle of attack is neglected. And the calculation of the bridge frequency is not precise enough due to the lack of video monitoring feature point data. These issues will be further explored and solved in the future studies.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Conflicts of Interest

The authors declare no conflicts of interest.

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